RAJALAKSHMI ENGINEERING COLLEGE

V Jikshitha 251801118-2025-AIDS V2

Started on	Saturday, 12 September 2026, 2:09 PM
State	Finished
Completed on	Saturday, 12 September 2026, 2:14 PM
Time taken	5 mins 47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>


#define MAX(a, b) ((a) > (b) ? (a) : (b))

int getLCSLength(const char *s1, const char *s2) {
    int m = strlen(s1);
    int n = strlen(s2);


    if (m < n) {
        const char *tempStr = s1;
        s1 = s2;
        s2 = tempStr;
        int tempLen = m;
        m = n;
        n = tempLen;
    }

    int *prev = (int *)calloc(n + 1, sizeof(int));
    int *curr = (int *)calloc(n + 1, sizeof(int));


    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                curr[j] = 1 + prev[j - 1];
            } else {
                curr[j] = MAX(prev[j], curr[j - 1]);
            }
        }

        for (int j = 0; j <= n; j++) {
            prev[j] = curr[j];
        }
    }

    int result = curr[n];

    free(prev);
    free(curr);
```

```
44
45      return result;
46  }
47
48  int main() {
49      char s1[1005];
50      char s2[1005];
51
52
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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